

Author Contributions Statement

Hardware-in-the-Loop Firmware Validation of the SEL-411L
Advanced Line Differential Relay via SAMBPS and IEC 61850 Sampled Values

Author contributions are reported using the CRediT (Contributor Roles Taxonomy) framework. Two authors are listed: **AVE** (Arun V. Eluvathingal) and **RKP** (R. K. Prasad, supervisor).

CRediT role	AVE	RKP
Conceptualisation	Lead	Supporting
Methodology	Lead	Supporting
Software	Lead	—
Validation	Lead	Supporting
Formal analysis	Lead	Supporting
Investigation	Lead	—
Resources	—	Lead
Writing – original draft	Lead	—
Writing – review & editing	Lead	Lead
Supervision	—	Lead

Detailed statement.

Arun V. Eluvathingal (AVE): Conceived and implemented the SAMBPS co-processor architecture on the SEL RTAC 3530. Derived Proposition 1 (critical-path timing bound) and Proposition 2 (float32 κ_n bound) with full perturbation-theory proofs. Designed and executed all 48 HIL scenarios; characterised deviations D3 and D4. Implemented IEC 61850 Sampled Value ingestion and GOOSE coordination on the RTAC. Wrote all sections of the manuscript.

R. K. Prasad (RKP): Provided research direction and laboratory resources (SEL-411L, SEL RTAC 3530, RTDS GPC-III hardware). Reviewed all technical propositions and validation results. Edited and approved the final manuscript.

Plain-text version (for journal submission portal):

AVE: Conceptualisation (Lead); Methodology (Lead); Software (Lead); Validation (Lead); Formal analysis (Lead); Investigation (Lead); Writing – original draft (Lead); Writing – review & editing (Lead).

RKP: Conceptualisation (Supporting); Methodology (Supporting); Validation (Supporting); Formal analysis (Supporting); Resources (Lead); Writing – review & editing (Lead); Supervision (Lead).